Project 3 — Example: Palmer penguins

Sample report for Project 3

Here I've done a sample Project 3 using the Palmer penguins TidyTuesday week (2020-07-28 on GitHub). Your project must use a different dataset.

Open sample in Colab · Sample one-page PDF

The sample notebook follows the Project 3 loading workflow: open the TidyTuesday readme, copy the `read_csv` URL, load with `library(tidyverse)`, then `glimpse()`. It includes all four univariate plots and all six bivariate plots. The one-page PDF shows the two figures I would submit plus findings in a student voice.

Introduction

How do flipper length, body mass, species, and island co-vary among Antarctic penguins in this sample? I loaded the CSV from TidyTuesday week 2020-07-28 with `read_csv()` (URL copied from that week's readme). Four focus variables: species (categorical), island (categorical), flipper length mm (numerical), body mass g (numerical). The full file has 344 rows; two are missing flipper length and body mass, so after a missingness check I analyze the 342 penguins with complete data on all four variables.

Univariate summaries (four variables — all four plots in notebook)

Variable	Type	Graph in notebook	Numeric fact
Species	Categorical	Bar chart	Gentoo: 124/342 36% of penguins
Island	Categorical	Bar chart	Biscoe: 168/342 49%
Flipper length	Numerical	Histogram	Mean 200.9 mm, SD 14.1 mm (median 197 mm, IQR 23 mm)
Body mass	Numerical	Histogram	Mean 4202 g, SD 802 g (median 4050 g, IQR 1200 g; right-skew)

Bivariate pairs (six pairs — all six plots in notebook)

Pair	Types	Graph + summary in notebook
Species × island	Cat × cat	100% stacked bar + two-way table (counts, row & column proportions)
Species × flipper length	Cat × num	Side-by-side boxplots + mean/SD/median/IQR per species (Gentoo 217 mm, Adelie 190 mm, Chinstrap 196 mm)
Species × body mass	Cat × num	Side-by-side boxplots + mean/SD per species (Gentoo 5076 g, Adelie 3701 g, Chinstrap 3733 g)
Island × flipper length	Cat × num	Side-by-side boxplots + mean/SD per island
Island × body mass	Cat × num	Side-by-side boxplots + mean/SD per island

Pair	Types	Graph + summary in notebook
Flipper length × body mass	Num × num	Scatterplot; Pearson r 0.87

Notable patterns: Gentoo have the longest flippers and heaviest bodies; Gentoo appear almost only on Biscoe; flipper length and body mass rise together linearly.

Two graphics chosen for the one-page summary

Figure 1: Side-by-side boxplots of flipper length by species (strong separation by species).
Figure 2: Scatterplot of flipper length vs body mass (clear positive linear trend).

Four numeric facts on the one-pager: e.g. Gentoo proportion; median flipper length; mean body mass; Pearson r for flipper vs body mass.

The other eight plots remain in the notebook only.

Conclusion (descriptive)

In this sample, species separates flipper length and body mass clearly, flipper length and body mass move together linearly, and Gentoo are concentrated on Biscoe. These are descriptions of the dataset, not claims about all penguins worldwide.

Runnable code for all ten plots: Open in Colab.